



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

THE DISCRETE FOURIER TRANSFORM

OBJECTIVES

- Understand the discrete Fourier transform.
- Know the properties of the Discrete Fourier transform
- Understand the Discrete cosine transform

1. Representation of Periodic Sequence: the Discrete Fourier Series
2. Properties of the Discrete Fourier Series
3. The Fourier Transform of a Periodic Signal
4. Fourier Representation of Finite-Duration Sequence: the Discrete Fourier Transform
5. Properties of the Discrete Fourier Transform

- Given a periodic sequence $\tilde{x}[n]$ with period N so that
- The Fourier series representation can be written as

$$\tilde{x}[n] = \tilde{x}[n + rN]$$
- Fourier series representation of continuous-time periodic signals require infinite many complex exponentials

$$\tilde{x}[n] = \frac{1}{N} \sum_{\underset{\text{red}}{k}} \tilde{X}[k] e^{j(2\pi/N)kn}$$

- Not that for discrete-time periodic signals we have

$$e^{j(2\pi/N)(k+mN)n} = e^{j(2\pi/N)kn} e^{j(2\pi mn)} = e^{j(2\pi/N)kn}$$

- Due to the periodicity of the complex exponential, we only need N exponentials for discrete time Fourier series

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

- No need

$$\tilde{x}[n] = \frac{1}{N} \sum_k \tilde{X}[k] e^{j(2\pi/N)kn}$$

- Discrete Fourier Series Pair

- A periodic sequence in terms of Fourier series coefficients

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

- To obtain the Fourier series coefficients, we multiply both sides by $e^{-j(2\pi/N)rn}$ for $0 \leq n \leq N-1$ and then sum both the sides, we obtain

$$\begin{aligned} \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}rn} &= \sum_{n=0}^{N-1} \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}(k) e^{j\frac{2\pi}{N}(k-r)n} \\ \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}rn} &= \sum_{k=0}^{N-1} \tilde{X}(k) \sum_{n=0}^{N-1} \frac{1}{N} e^{j\frac{2\pi}{N}(k-r)n} \end{aligned}$$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

$$\sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}rn} = \sum_{k=0}^{N-1} \tilde{X}(k) \sum_{n=0}^{N-1} \frac{1}{N} e^{j\frac{2\pi}{N}(k-r)n}$$

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-r)n} = \begin{cases} 1, & k - r = mN, \text{ } m \text{ an integer} \\ 0, & \text{otherwise} \end{cases}$$

$$\sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}rn} = \tilde{X}(r)$$

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}kn}$$

- Representation of Periodic Sequence: the Discrete Fourier Series

- a periodic sequence $\tilde{x}[n]$ with period N ,

- The Fourier series coefficients of $\tilde{x}[n]$ is

$$\tilde{x}[n] = \tilde{x}[n + rN] \text{ for any integer } r$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn}$$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn}$$

◆ The sequence $\tilde{X}[k]$ is periodic with period N

$$\tilde{X}[0] = \tilde{X}[N], \quad \tilde{X}[1] = \tilde{X}[N+1]$$

$$\begin{aligned} \tilde{X}[k+N] &= \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)(k+N)n} \\ &= \left(\sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn} \right) e^{-j2\pi n} = \tilde{X}[k] \end{aligned}$$

- Discrete Fourier Series (DFS)

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn} \quad \blacksquare \text{ Let } W_N = e^{-j(2\pi/N)}$$

◆ Analysis equation:

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn}$$

◆ Synthesis equation:

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn}$$

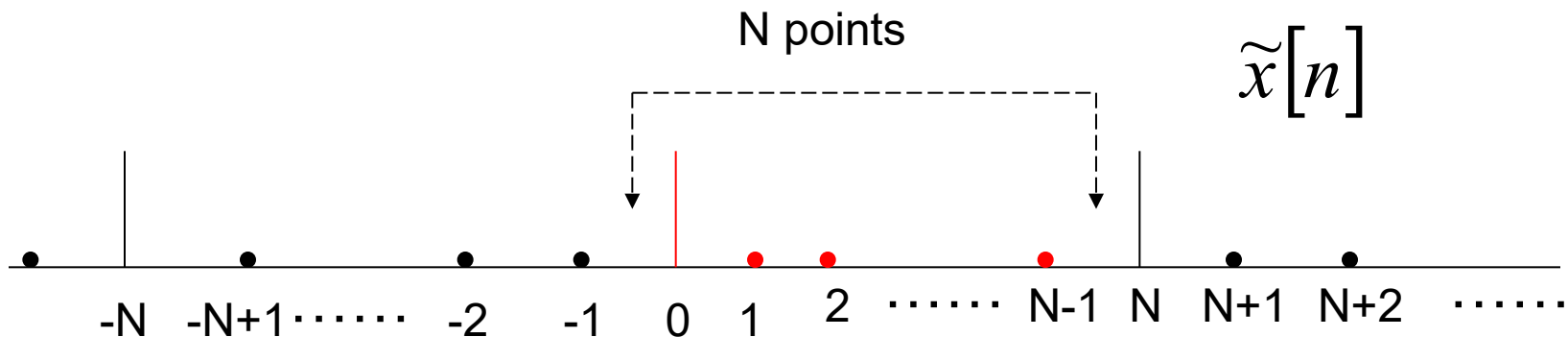
$$\tilde{x}[n] \overset{DFS}{\longleftrightarrow} \tilde{X}[k]$$

discrete $\overset{F}{\longleftrightarrow}$ *periodic*

periodic $\overset{F}{\longleftrightarrow}$ *discrete*

- Example: DFS of an impulse train
- Consider the periodic impulse train

$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} \delta[n - rN] = \begin{cases} 1, & n = rN, r \text{ is any integer} \\ 0, & \text{otherwise} \end{cases}$$

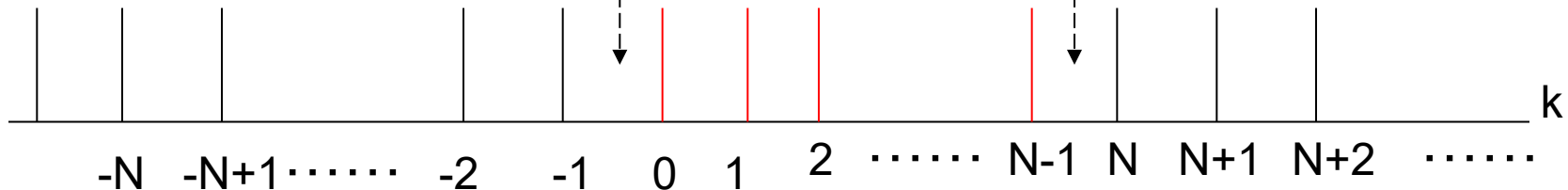


$$\tilde{X}[k] = \sum_{n=0}^{N-1} \delta[n] W_N^{kn} = W_N^0 = 1$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \delta[n] W_N^{kn} = W_N^0 = 1$$

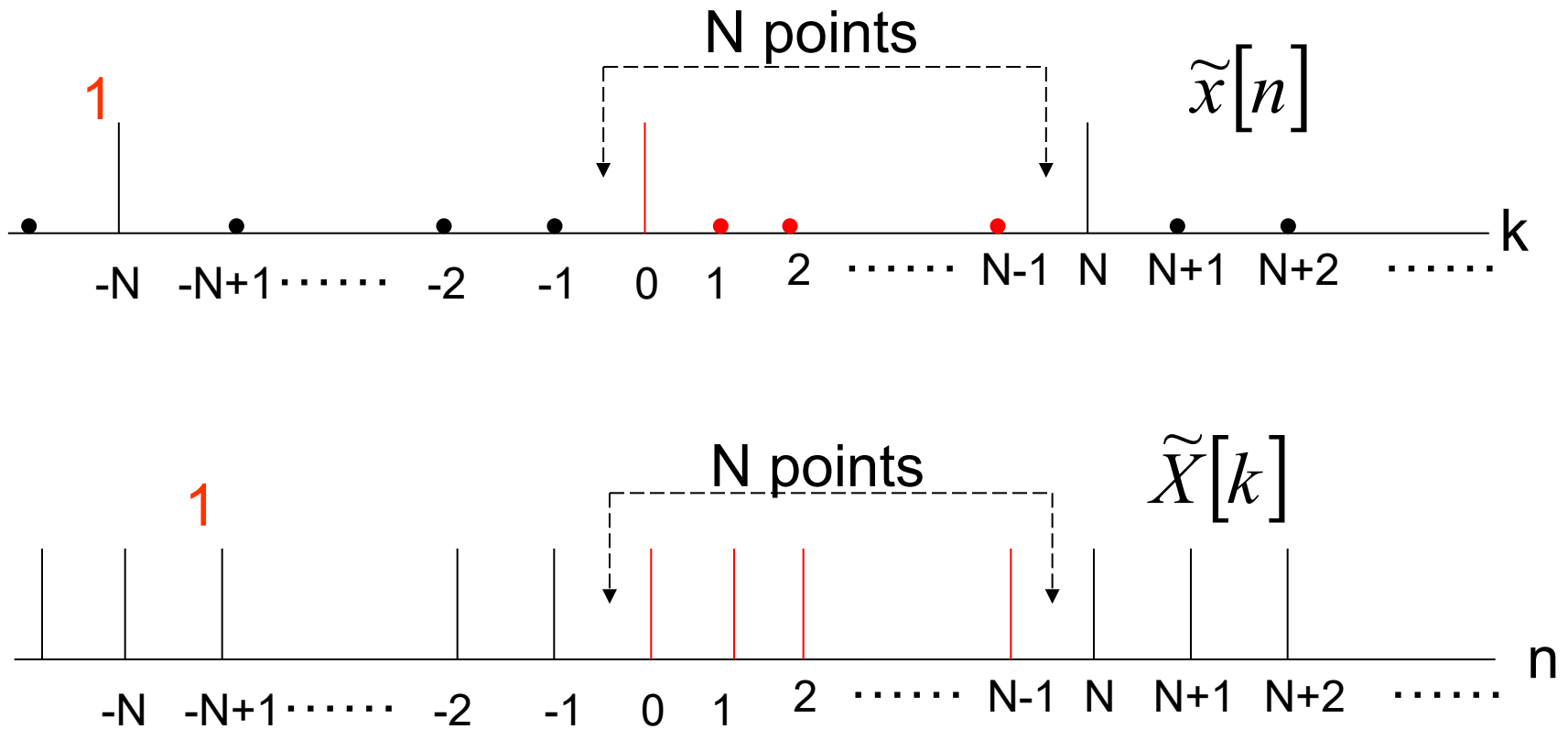
N points

$\tilde{X}[k]$

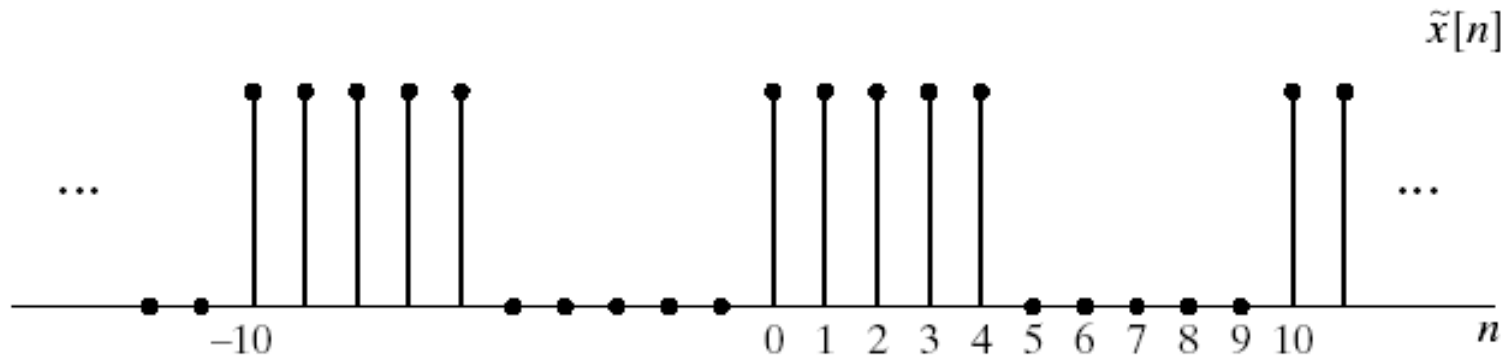


$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} \delta[n - rN] = \begin{cases} 1, & n = rN, r \text{ is any integer} \\ 0, & \text{otherwise} \end{cases}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} e^{j(2\pi/N)kn}$$



- Example: The Discrete Fourier Series of a Periodic Rectangular Pulse Train
 - Periodic sequence with period $N=10$

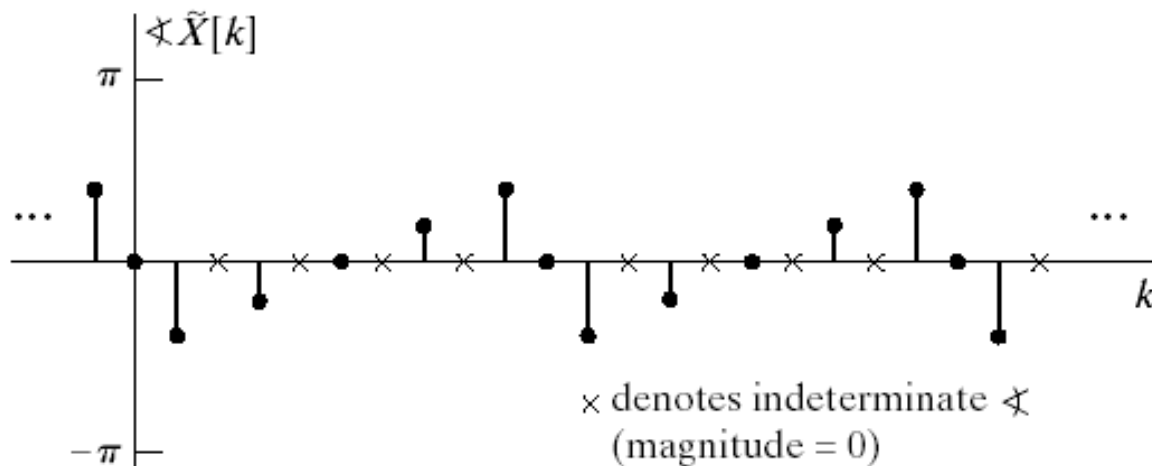


$$\begin{aligned}\tilde{X}[k] &= \sum_{n=0}^4 W_{10}^{kn} = \sum_{n=0}^4 e^{-j(2\pi/10)kn} = \frac{1 - W_{10}^{5k}}{1 - W_{10}^k} \\ &= e^{-(4\pi k/10)} \frac{\sin(\pi k/2)}{\sin(\pi k/10)}\end{aligned}$$

magnitude

$$\tilde{X}[k] = e^{-(4\pi k/10)} \frac{\sin(\pi k/2)}{\sin(\pi k/10)}$$

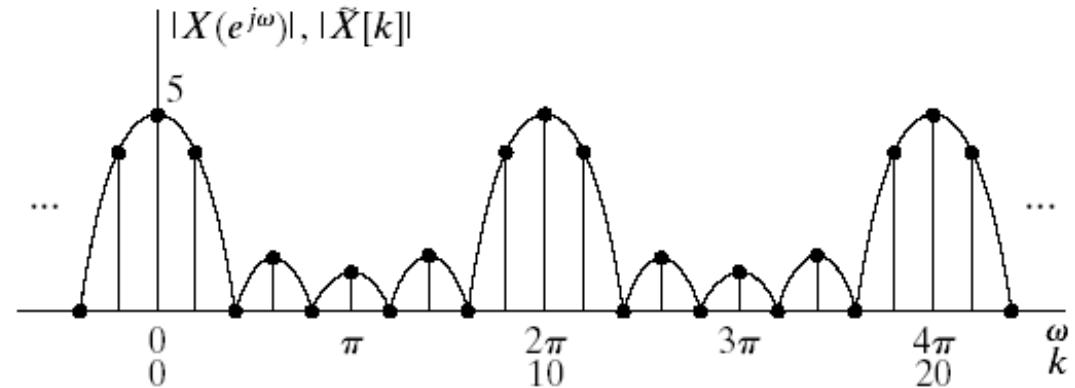
phase



THE DISCRETE FOURIER SERIES

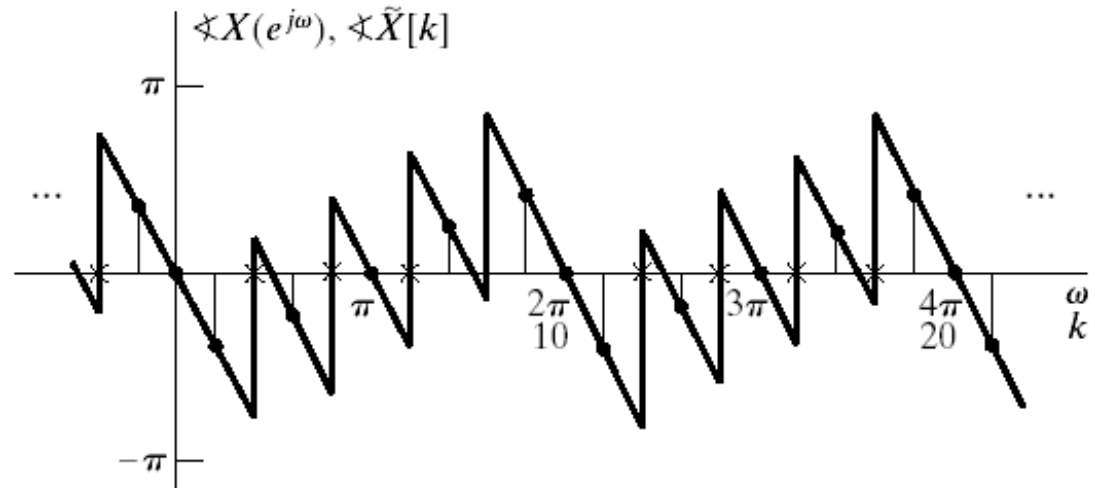
magnitude

$$\tilde{X}[k] = e^{-(4\pi k/10)} \frac{\sin(\pi k/2)}{\sin(\pi k/10)}$$



(a)

phase



1. Representation of Periodic Sequence: the Discrete Fourier Series
2. Properties of the Discrete Fourier Series
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4. Fourier Representation of Finite-Duration Sequence: the Discrete Fourier Transform
5. Properties of the Discrete Fourier Transform

- **Linearity**: two periodic sequence, both with period N

$$\tilde{x}_1[n] \overset{DFS}{\longleftrightarrow} \tilde{X}_1[k]$$

$$\tilde{x}_2[n] \overset{DFS}{\longleftrightarrow} \tilde{X}_2[k]$$

$$a\tilde{x}_1[n] + b\tilde{x}_2[n] \overset{DFS}{\longleftrightarrow} a\tilde{X}_1[k] + b\tilde{X}_2[k]$$

- Shift of a sequence

$$\tilde{x}[n] \overset{DFS}{\longleftrightarrow} \tilde{X}[k]$$

$$\tilde{x}[n-m] \overset{DFS}{\longleftrightarrow} W_N^{km} \tilde{X}[k]$$

$$W_N^{-nl} \tilde{x}[n] \overset{DFS}{\longleftrightarrow} \tilde{X}[k-l]$$

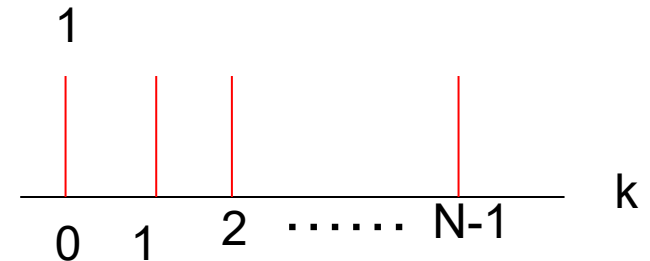
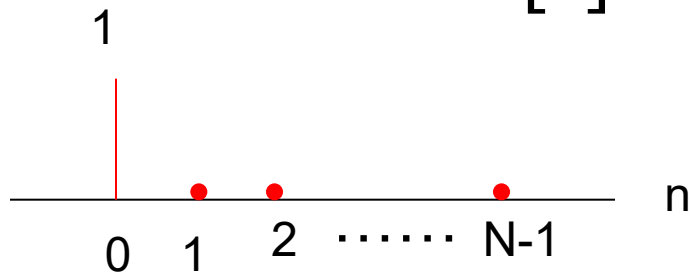
• Duality

$$\tilde{x}[n] \xleftrightarrow{DFS} \tilde{X}[k]$$

$$\tilde{X}[n] \xleftrightarrow{DFS} N\tilde{x}[-k]$$

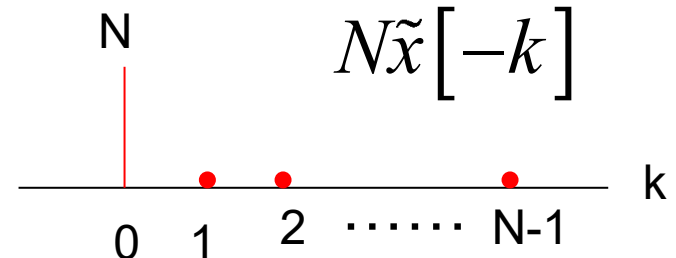
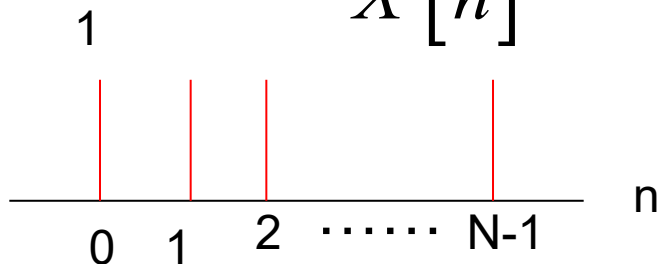
$$\tilde{x}[n]$$

$$\tilde{X}[k]$$



$$\tilde{X}[n]$$

$$N\tilde{x}[-k]$$



1. Representation of Periodic Sequence: the Discrete Fourier Series
2. Properties of the Discrete Fourier Series
- 3. The Fourier Transform of a Periodic Signal**
4. Fourier Representation of Finite-Duration Sequence: the Discrete Fourier Transform
5. Properties of the Discrete Fourier Transform

- Periodic sequences are neither absolutely summable nor square summable, hence they don't have a strict Fourier Transform

$$x[n] = 1 \text{ for all } n \quad \xleftrightarrow{F} \quad X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi \delta(\omega + 2\pi r)$$

$$x[n] = e^{j\omega_0 n} \quad \xleftrightarrow{F} \quad X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi r)$$

$$x[n] = \sum_k a_k e^{j\omega_k n} \quad \xleftrightarrow{F} \quad X(e^{j\omega}) = \sum_{r=-\infty}^{\infty} \sum_k 2\pi a_k \delta(\omega - \omega_k + 2\pi r)$$

- We can represent Periodic sequences as sums of complex exponentials: DFS
- We can combine DFS and the Fourier transform
- Fourier transform of periodic sequences
- Periodic impulse train with values proportional to DFS coefficients

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j(2\pi/N)kn}$$

$$\tilde{X}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \tilde{X}[k] \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\tilde{X}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \tilde{X}[k] \delta\left(\omega - \frac{2\pi k}{N}\right)$$

This is periodic with 2π since DFS is periodic

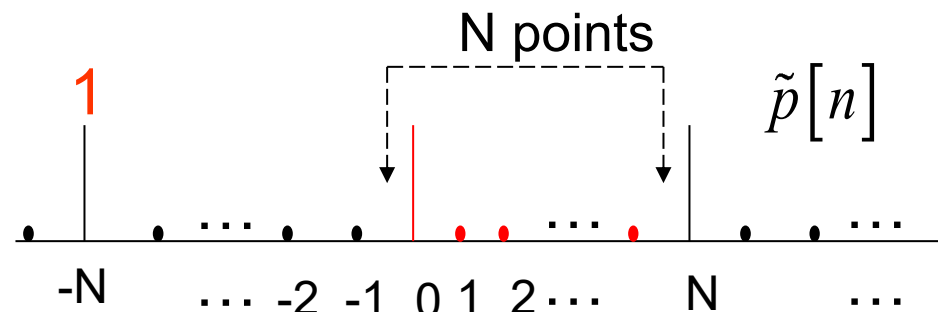
• The inverse transform can be written as

$$\begin{aligned} \frac{1}{2\pi} \int_{0-\varepsilon}^{2\pi-\varepsilon} \tilde{X}(e^{j\omega}) e^{j\omega n} d\omega &= \frac{1}{2\pi} \int_{0-\varepsilon}^{2\pi-\varepsilon} \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \tilde{X}[k] \delta\left(\omega - \frac{2\pi k}{N}\right) e^{j\omega n} d\omega \\ &= \frac{1}{N} \sum_{k=-\infty}^{\infty} \tilde{X}[k] \int_{0-\varepsilon}^{2\pi-\varepsilon} \delta\left(\omega - \frac{2\pi k}{N}\right) e^{j\omega n} d\omega = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi k}{N}n} \\ &= \tilde{x}[n] \end{aligned}$$

- Example: Fourier Transform of a periodic impulse train

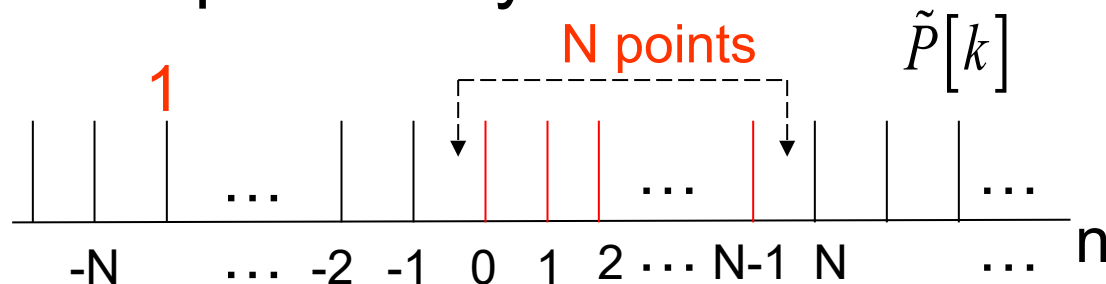
- Consider the periodic impulse train

$$\tilde{p}[n] = \sum_{r=-\infty}^{\infty} \delta[n - rN]$$



- ◆ The DFS was calculated previously to be

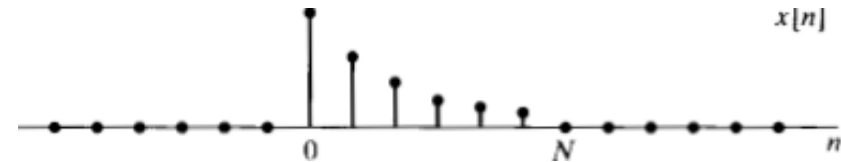
$$\tilde{P}[k] = 1 \text{ for all } k$$



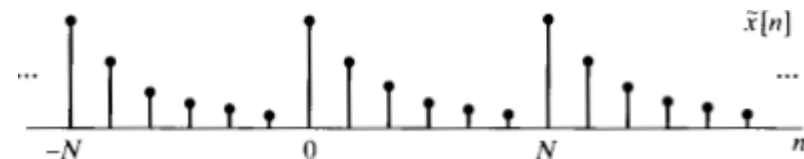
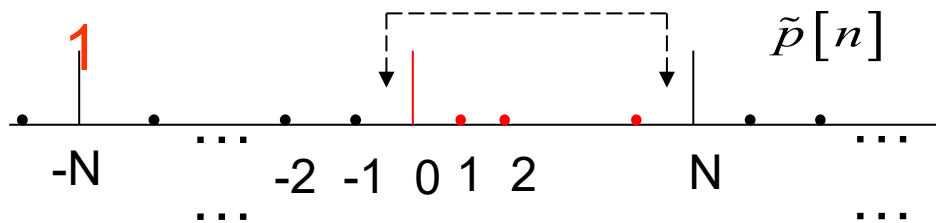
◆ Therefore the Fourier transform is

$$\tilde{P}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \delta\left(\omega - \frac{2\pi k}{N}\right)$$

- Relation between Finite-length and Periodic Signals
 - Consider finite length signal $x[n]$ spanning from 0 to $N-1$



- Convolve with periodic impulse train



$$\tilde{x}[n] = x[n] * \tilde{p}[n] = x[n] * \sum_{r=-\infty}^{\infty} \delta[n - rN] = \sum_{r=-\infty}^{\infty} x[n - rN]$$

- The Fourier transform of the periodic sequence is

$$\tilde{X}(e^{j\omega}) = X(e^{j\omega}) \tilde{P}(e^{j\omega}) = X(e^{j\omega}) \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\tilde{X}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} X\left(e^{j\frac{2\pi k}{N}}\right) \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\tilde{X}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} \tilde{X}[k] \delta\left(\omega - \frac{2\pi k}{N}\right)$$

$$\tilde{X}(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \frac{2\pi}{N} X\left(e^{j\frac{2\pi k}{N}}\right) \delta\left(\omega - \frac{2\pi k}{N}\right)$$

- ◆ This implies that $\tilde{X}[k] = X\left(e^{j\frac{2\pi k}{N}}\right) = X\left(e^{j\omega}\right)\Big|_{\omega=\frac{2\pi k}{N}}$
- ◆ DFS coefficients of a periodic signal can be thought as **equally spaced samples** of the Fourier transform of one period

- If $\tilde{x}[n]$ is periodic with period N , the DFS are

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j(2\pi/N)kn}$$

- If $x[n] = \tilde{x}[n]$ for $0 \leq n \leq N-1$ and $x[n] = 0$ otherwise then

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n] e^{-j\omega n} = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\omega n}$$

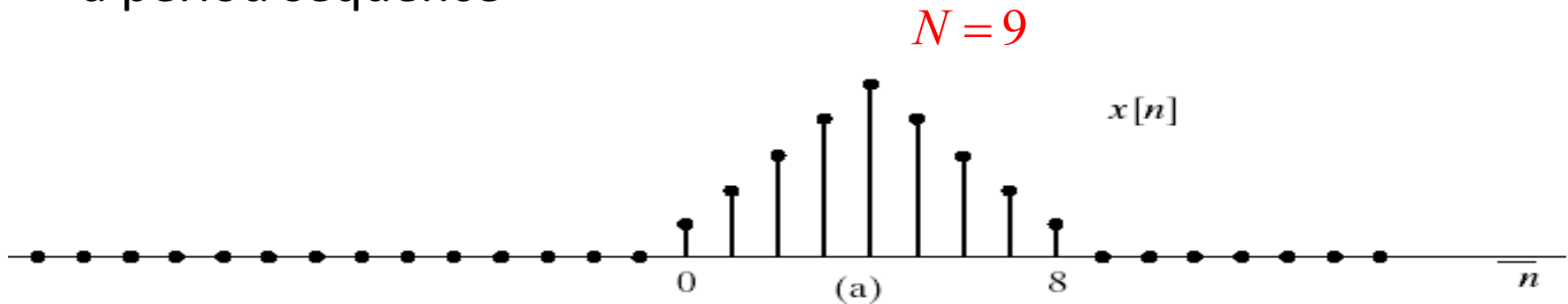
$$\tilde{X}[k] = X(e^{j\omega}) \Big|_{\omega=2\pi k/N}$$

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- Consider a finite-length sequence $x[n]$ of length N samples such that $x[n] = 0$ outside the range

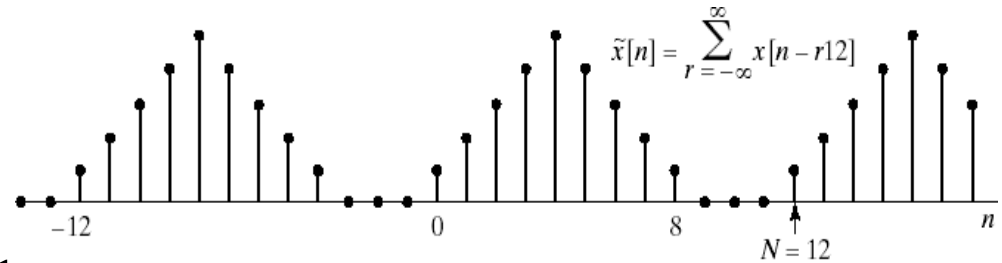
$$0 \leq n \leq N-1$$

- To each finite-length sequence of length N , we can associate a period sequence



$$\tilde{x}[n] = \sum_{r=-\infty}^{\infty} x[n - rN]$$

$$x[n] = \begin{cases} \tilde{x}[n], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$



$$\tilde{x}[n] = x[(n \bmod N)] = x\left[\left((n)\right)_N\right]$$

- For $\tilde{x}[n]$, the DFS is $\tilde{X}[k]$ with period N

•

- The **Discrete Fourier Transform** of $x[n]$ is

$$X[k] = \begin{cases} \tilde{X}[k], & 0 \leq k \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$\tilde{X}[k] = X[(k \bmod N)] = X[(k)_N]$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn}$$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn}$$

$$X[k] = \begin{cases} \sum_{n=0}^{N-1} x[n] W_N^{kn}, & 0 \leq k \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$x[n] = \begin{cases} \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

- Analysis equation

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$

- Synthesis equation

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

$$x[n] \xleftrightarrow{DFT} X[k]$$

THE DISCRETE FOURIER TRANSFORM

	Time	Frequency
Fourier transform (FT)	continuous	continuous
Fourier series (FS)	continuous periodic	Continuous impulse train
Discrete-time Fourier transform (DTFT)	discrete	continuous periodic
Discrete Fourier series (DFS)	discrete periodic	continuous impulse train, periodic
Discrete Fourier transform (DFT)	discrete	discrete

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•Linearity

$$ax_1[n] + bx_2[n] \xleftrightarrow{DFT} aX_1[k] + bX_2[k]$$

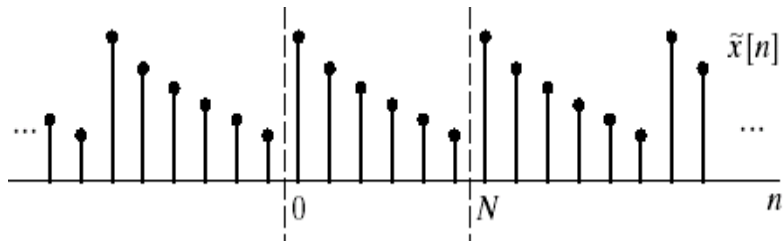
◆ If $x_1[n]$ has length N_1 and $x_2[n]$ has length N_2 ,

$$N_3 = \max(N_1, N_2)$$

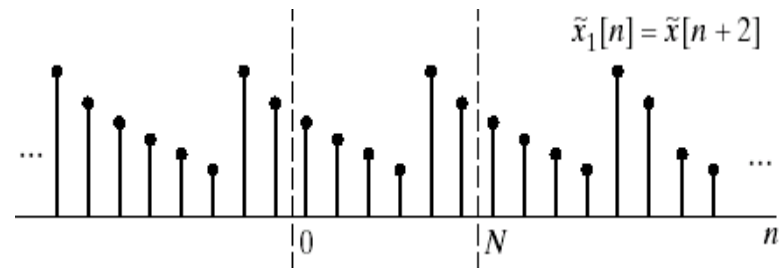
$$X_1[k] = \sum_{n=0}^{N_3-1} x_1[n] W_{N_3}^{kn}, \quad 0 \leq k \leq N_3 - 1$$

$$X_2[k] = \sum_{n=0}^{N_3-1} x_2[n] W_{N_3}^{kn}, \quad 0 \leq k \leq N_3 - 1$$

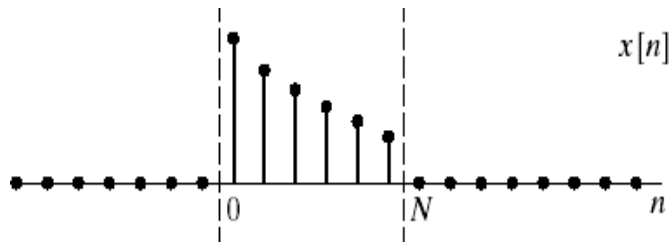
- Circular Shift of a Sequence



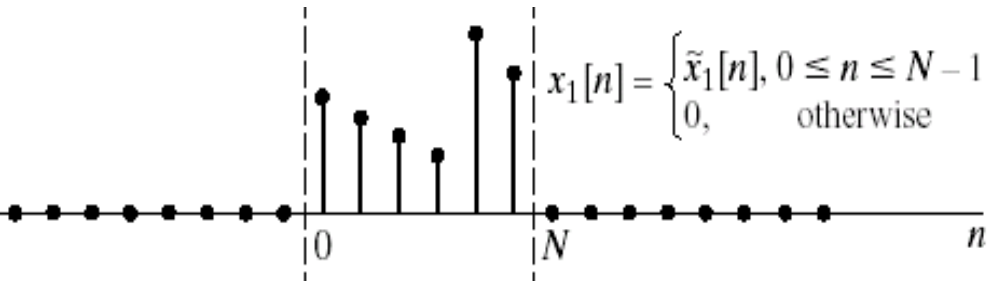
$$\tilde{x}[n] \xleftrightarrow{DFS} \tilde{X}[k]$$



$$\tilde{x}[n-m] \xleftrightarrow{DFS} e^{-j(2\pi k/N)m} \tilde{X}[k]$$



$$x[n], \quad 0 \leq n \leq N-1 \xleftrightarrow{DFT} X[k]$$



$$x\left[\left((n-m)\right)_N\right], \quad 0 \leq n \leq N-1 \quad \xleftrightarrow{DFT} \quad e^{-j(2\pi k/N)m} X[k]$$

$$x[n] \leftrightarrow X(e^{j\omega})$$

$$x[n-m] \leftrightarrow e^{-j\omega m} X(e^{j\omega})$$

$$\tilde{x}[n] \xleftrightarrow{DFS} \tilde{X}[k]$$

$$\tilde{x}[n-m] \xleftrightarrow{DFS} e^{-j(2\pi k/N)m} \tilde{X}[k]$$

$$x[n], \quad 0 \leq n \leq N-1 \quad \xleftrightarrow{DFT} \quad X[k]$$

$$x_1[n], \quad 0 \leq n \leq N-1 \quad \xleftrightarrow{DFT} \quad X_1[k] = e^{-j(2\pi k/N)m} X[k]$$

$$\tilde{x}[n] = x\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}[k] = X\left[\left((k)\right)_N\right]$$

$$\tilde{x}_1[n] = x_1\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}_1[k] = X_1\left[\left((k)\right)_N\right]$$

$$\tilde{x}[n] = x\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}[k] = X\left[\left((k)\right)_N\right]$$

$$\tilde{x}_1[n] = x_1\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}_1[k] = X_1\left[\left((k)\right)_N\right]$$

$$X_1[k] = e^{-j(2\pi k/N)m} X[k]$$

$$\begin{aligned} \tilde{X}_1[k] &= e^{-j\left(2\pi \left((k)\right)_N / N\right)m} X\left[\left((k)\right)_N\right] \\ &= e^{-j(2\pi k/N)m} X\left[\left((k)\right)_N\right] = e^{-j(2\pi k/N)m} \tilde{X}[k] \end{aligned}$$

$$\tilde{x}_1[n] = x\left[\left((n-m)\right)_N\right] = \tilde{x}[n-m] \xleftrightarrow{DFS} e^{-j(2\pi k/N)m} \tilde{X}[k]$$

$$X_1[k] = e^{-j(2\pi k/N)m} X[k]$$

$$\tilde{x}_1[n] = x\left[\left((n-m)\right)_N\right] = \tilde{x}[n-m] \xleftrightarrow{DFS} e^{-j(2\pi k/N)m} \tilde{X}[k]$$

$$x_1[n] = \begin{cases} \tilde{x}_1[n] = x\left[\left((n-m)\right)_N\right], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$x[n], \quad 0 \leq n \leq N-1 \quad \xleftrightarrow{DFT} \quad X[k]$$

$$x\left[\left((n-m)\right)_N\right], \quad 0 \leq n \leq N-1 \quad \xleftrightarrow{DFT} \quad e^{-j(2\pi k/N)m} X[k]$$

• Duality

$$\tilde{x}[n] \xleftrightarrow{DFS} \tilde{X}[k]$$

$$\tilde{x}[n] = x\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}[k] = X\left[\left((k)\right)_N\right]$$

$$\tilde{X}[n] \xleftrightarrow{DFS} N\tilde{x}[-k] \quad \tilde{x}_1[n] = \tilde{X}[n] \quad \tilde{X}_1[k] = N\tilde{x}[-k]$$

$$x_1[n] = X[n]$$

$$X_1[k] = \begin{cases} N\tilde{x}[-k] = Nx\left[\left((-k)\right)_N\right], & 0 \leq k \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

$$x[n] \xleftrightarrow{DFT} X[k] \quad X[n] \xleftrightarrow{DFT} Nx\left[\left((-k)\right)_N\right], \quad 0 \leq k \leq N-1$$

•Symmetry Properties

$$\tilde{x}[n] = x\left[\left((n)\right)_N\right] \xleftrightarrow{DFS} \tilde{X}[k] = X\left[\left((k)\right)_N\right]$$

$$x^*[n] \xleftrightarrow{DFT} X^*\left[\left((-k)\right)_N\right], \quad 0 \leq n \leq N-1$$

$$x^*\left[\left((-n)\right)_N\right] \xleftrightarrow{DFT} X^*[k], \quad 0 \leq n \leq N-1$$

$$\tilde{x}_e[n] = \frac{1}{2} \left\{ \tilde{x}[n] + \tilde{x}^*[-n] \right\}$$

$$\tilde{x}_o[n] = \frac{1}{2} \left\{ \tilde{x}[n] - \tilde{x}^*[-n] \right\}$$

$$x_{ep}[n] = \tilde{x}_e[n] = \frac{1}{2} \left\{ x\left[\left((n)\right)_N\right] + x^*\left[\left((-n)\right)_N\right] \right\}, 0 \leq n \leq N-1$$

$$x_{op}[n] = \tilde{x}_o[n] = \frac{1}{2} \left\{ x\left[\left((n)\right)_N\right] - x^*\left[\left((-n)\right)_N\right] \right\}, 0 \leq n \leq N-1$$

$$\text{for } 0 \leq n \leq N-1, \quad \left((-n)\right)_N = N-n, \quad \left((n)\right)_N = n$$

$$x_{ep}[n] = \frac{1}{2} \left\{ x[n] + x^*[N-n] \right\}, \quad 0 \leq n \leq N-1$$

$$x_{ep}[0] = \frac{1}{2} \left\{ x[0] + x^*[N] \right\} = \frac{1}{2} \left\{ x[0] + x^*[0] \right\} = \text{Re}\{x[0]\}$$

$$x_{op}[n] = \frac{1}{2} \left\{ x[n] - x^*[N-n] \right\}, \quad 0 \leq n \leq N-1$$

$$x_{op}[0] = \frac{1}{2} \left\{ x[0] - x^*[0] \right\} = j \text{Im}\{x[0]\}$$

$$x_{ep}[n] = \frac{1}{2} \{x[n] + x^*[N-n]\}, \quad 0 \leq n \leq N-1$$

$$x_{op}[n] = \frac{1}{2} \{x[n] - x^*[N-n]\}, \quad 0 \leq n \leq N-1$$

$$x_e[n] = \frac{1}{2} \{x[n] + x^*[-n]\} \quad x_o[n] = \frac{1}{2} \{x[n] - x^*[-n]\}$$

$$x_{ep}[n] = \{x_e[n] + x_e[n-N]\}, \quad 0 \leq n \leq N-1$$

$$x_{op}[n] = \{x_o[n] + x_o[n-N]\}, \quad 0 \leq n \leq N-1$$

$$\tilde{x}[n] = \tilde{x}_e[n] + \tilde{x}_o[n]$$

$$x[n] = \tilde{x}[n] = \tilde{x}_e[n] + \tilde{x}_o[n], \quad 0 \leq n \leq N-1$$

$$x_{ep}[n] = \tilde{x}_e[n], \quad 0 \leq n \leq N-1$$

$$x_{op}[n] = \tilde{x}_o[n], \quad 0 \leq n \leq N-1$$

$$x[n] = x_{ep}[n] + x_{op}[n]$$

$$\text{Re}\{x[n]\} \xleftrightarrow{DFT} X_{ep}[k]$$

$$j \text{Im}\{x[n]\} \xleftrightarrow{DFT} X_{op}[k]$$

$$x_{ep}[n] \xleftrightarrow{DFT} \text{Re}\{X[k]\}$$

$$x_{op}[n] \xleftrightarrow{DFT} j \text{Im}\{X[k]\}$$

5. $x[((n - m))_N]$

$W_N^{km} X[k]$

6. $W_N^{-\ell n} x[n]$

$X[((k - \ell))_N]$

7. $\sum_{m=0}^{N-1} x_1(m) x_2[((n - m))_N]$

$X_1[k] X_2[k]$

8. $x_1[n] x_2[n]$

$\frac{1}{N} \sum_{\ell=0}^{N-1} X_1(\ell) X_2[((k - \ell))_N]$

9. $x^*[n]$

$X^*[((-k))_N]$

10. $x^*[((-n))_N]$

$X^*[k]$

11. $\mathcal{Re}\{x[n]\}$

$X_{\text{ep}}[k] = \frac{1}{2} \{X[((k))_N] + X^*[((-k))_N]\}$

12. $j\mathcal{Im}\{x[n]\}$

$X_{\text{op}}[k] = \frac{1}{2} \{X[((k))_N] - X^*[((-k))_N]\}$

13. $x_{\text{ep}}[n] = \frac{1}{2} \{x[n] + x^*[(((-n))_N)]\}$

$\mathcal{Re}\{X[k]\}$

14. $x_{\text{op}}[n] = \frac{1}{2} \{x[n] - x^*[(((-n))_N)]\}$

$j\mathcal{Im}\{X[k]\}$

Properties 15–17 apply only when $x[n]$ is real.

15. Symmetry properties

$$\begin{cases} X[k] = X^*[((-k))_N] \\ \mathcal{Re}\{X[k]\} = \mathcal{Re}\{X^*[((-k))_N]\} \\ \mathcal{Im}\{X[k]\} = -\mathcal{Im}\{X^*[((-k))_N]\} \\ |X[k]| = |X^*[((-k))_N]| \\ \angle\{X[k]\} = -\angle\{X^*[((-k))_N]\} \end{cases}$$

•Circular Convolution

$$\sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n-m] \xleftrightarrow{DFS} \tilde{X}_1[k] \tilde{X}_2[k]$$

◆ For two **finite-duration** sequences $x_1[n]$ and $x_2[n]$, both of length N , with DFTs $X_1[k]$ and $X_2[k]$

$$X_3[k] = X_1[k] X_2[k]$$

$$x_3[n] = \sum_{m=0}^{N-1} \tilde{x}_1[m] \tilde{x}_2[n-m], \quad 0 \leq k \leq N-1$$

$$x_3[n] = \sum_{m=0}^{N-1} x_1\left[\left((m)\right)_N\right] x_2\left[\left((n-m)\right)_N\right], \quad 0 \leq k \leq N-1$$

$$x_3[n] = \sum_{m=0}^{N-1} x_1\left[\left((m)\right)_N\right] x_2\left[\left((n-m)\right)_N\right], \quad 0 \leq k \leq N-1$$

$$= \sum_{m=0}^{N-1} x_1[m] x_2\left[\left((n-m)\right)_N\right], \quad 0 \leq k \leq N-1$$

since $\left((m)\right)_N = m$, for $0 \leq k \leq N-1$

$$\begin{aligned} x_3[n] &= \sum_{m=0}^{N-1} x_1[m] x_2\left[\left((n-m)\right)_N\right] \\ &= x_1[n] \textcircled{N} x_2[n] \end{aligned}$$

$$\begin{aligned}
 x_3[n] &= \sum_{m=0}^{N-1} x_1[m] x_2\left[\left((n-m)\right)_N\right] \\
 &= x_1[n] \circledast x_2[n] = x_2[n] \circledast x_1[n] \\
 &= \sum_{m=0}^{N-1} x_2[m] x_1\left[\left((n-m)\right)_N\right]
 \end{aligned}$$

$$\overset{DFT}{x_3[n]} \longleftrightarrow X_3[k] = X_1[k] X_2[k], \quad 0 \leq k \leq N-1$$

$$x_1[n] \circledcirc x_2[n] \xleftrightarrow{DFT} X_1[k] X_2[k]$$

if $x_3[n] = x_1[n] x_2[n]$

$$X_3[k] = \frac{1}{N} \sum_{l=0}^{N-1} X_1[l] X_2[(k-l)_N]$$

$$x_1[n] x_2[n] \xleftrightarrow{DFT} \frac{1}{N} X_1[k] \circledcirc X_2[k]$$

- This chapter we have learnt :
 - Discrete Fourier Transform (DFT) for **finite duration** sequence
 - DFT is a **sequence** rather than a function of a continuous variable
 - DFT corresponds to **sample**, equally spaced in frequency, of the Fourier transform of the signal.
 - The Fourier series representation of the periodic sequence corresponds to the DFT of the finite-length sequence.



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THANK YOU
